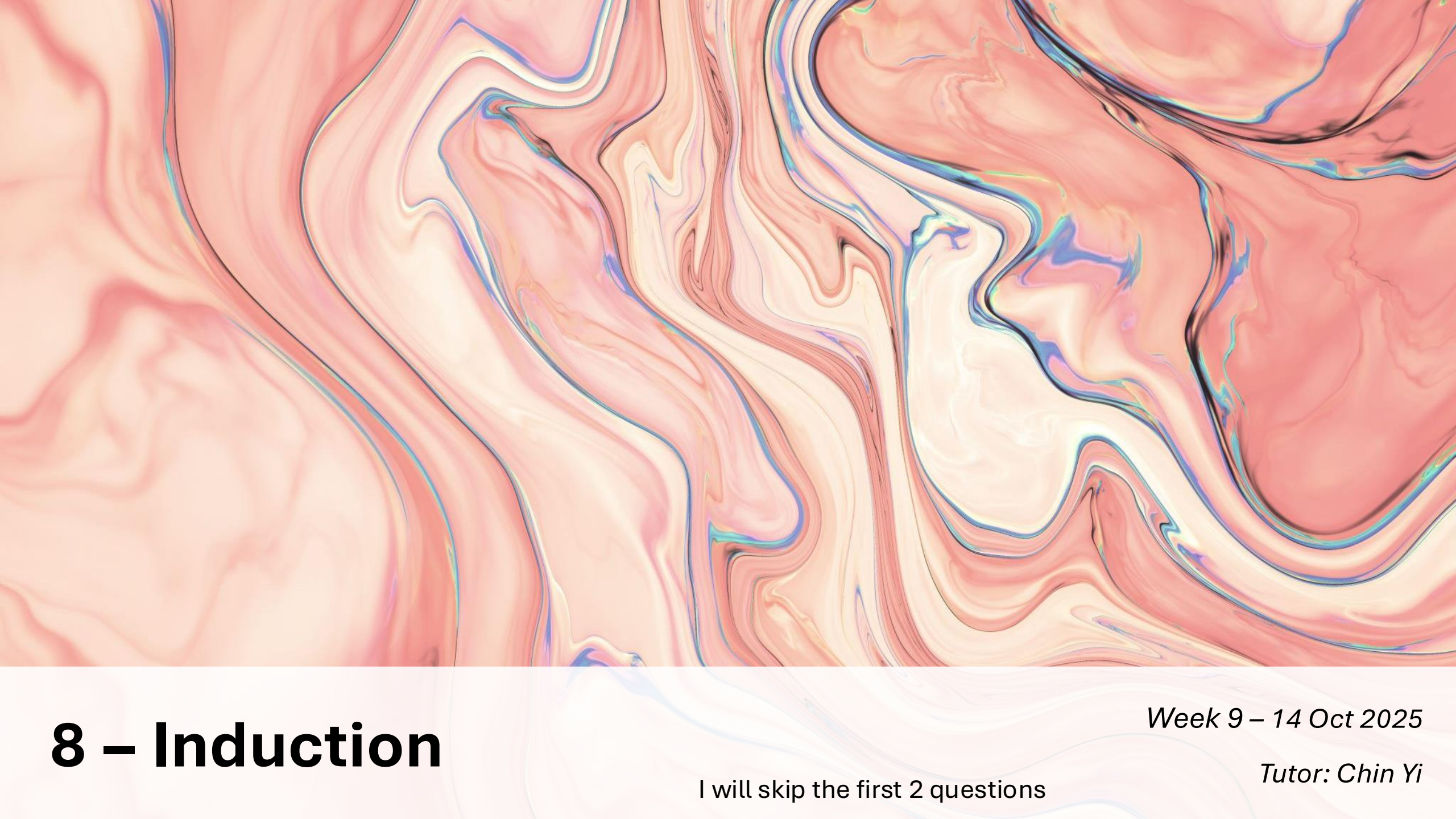




8 – Induction

Week 9 – 14 Oct 2025

Tutor: Chin Yi

I will skip the first 2 questions

Cheat sheet

How to do induction questions?

- We have $P(n)$
- Base case:
 - Show that $P(0)$ is valid
- Inductive step:
 - **Suppose $P(k)$ is true**
 - **WTS: $P(k+1)$ is true**

Ex. 5.2.14. Prove the following statement by mathematical induction.

$$\sum_{i=1}^{n+1} i \cdot 2^i = n \cdot 2^{n+2} + 2, \quad \text{for all integers } n \geq 0.$$

Ex. 5.3.11. Prove the following statement by mathematical induction.

$$3^{2n} - 1 \text{ is divisible by 8, for each integer } n \geq 0.$$

Ex. 5.3.12. Prove the following statement by mathematical induction.

$$7^n - 2^n \text{ is divisible by 5, for each integer } n \geq 0.$$

Ex. 5.3.20. Prove the following statement by mathematical induction.

$$2^n < (n + 2)!, \text{ for all integers } n \geq 0.$$

Ex. 5.3.22. Prove the following statement by mathematical induction.

$$1 + nx \leq (1 + x)^n, \text{ for all real numbers } x > -1 \text{ and integers } n \geq 2.$$

Week 10 – Tutorial 9 e-Learning



<https://youtu.be/SM7f9dC3nL0>